Local lab setup

If IT support hasn't set up an online server you can still start working by following this small guide to setting up locally. Note however that IT support do not have any obligation to help and TA's are no experts in docker and may not be able to help.

Setting up the server locally

1. Download and install Docker, follow the guide that corresponds to your OS here Link
2. Install PSQL, Link
    a. Installing PSQL might be easier using a package manager such as Homebrew for Mac.
3. Follow the steps below. We've tested the instructions for Mac & Windows with limited issues. However, as always when installing software, you might run into some issues. In that case, Google is your friend :)

Setup the the docker server:

- Open up a standard terminal session, and make sure that both Docker and PSQL is correctly configured, e.g. through:
    - `docker --version`
    - `psql --version`
        - You might need to set the Path variable to include the lib & bin folder of your PSQL installation. This was required in our case on a Windows machine.
- **Make sure the docker app is running**
- Pull PSQL to docker:
    - `docker pull postgres`
- Create the PSQL database, remember to replace the name, password, username and database name:
    - `docker run -d --name <Insert any name> -e POSTGRES_PASSWORD=<Insert password> -e POSTGRES_USER=<Insert username> -e POSTGRES_DB=<Insert database name> -p 5432:5432 postgres`
    - If everything went smoothly, you will see the docker instance in the docker app. From here, you can later stop/resume the process whenever you want
- Connect to the database:

        - `psql -h localhost -p 5432 -U <Insert username> -d <Insert database name>`
        - You will be prompted for the password you provided earlier upon executing this command
        - In case you are denied access despite entering the correct password, you might need to stop all running PSQL instances on your device (if there are any). Please see Link
- Check the current tables in the database (should be empty)
    - `\d`

You can now execute CREATE TABLE, INSERT and query the database from here.

You can also run .psql files in case you have saved a file with your CREATE TABLE and INSERT statements. For instance, if you have a file `create.psql`, you can execute all statements by running the following, assuming you `cd` into the directory containing the file:

- `psql -h localhost -U <Insert username> -d <Insert database name> -f create.psql`